

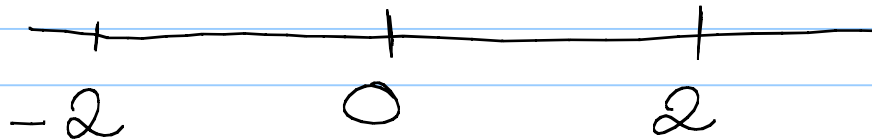
Unit 1.2 Absolute Value

Note Title

2013/02/11

Stewart p A6 - A8

- Notation: The absolute number of x is denoted by $|x|$.
- Geometric meaning: Distance from x to origin (0) on the real number line. It is always positive!



$$|2| = 2 \quad \text{and} \quad |-2| = 2$$

Definition of absolute value

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Example Express the following without using the absolute value symbol

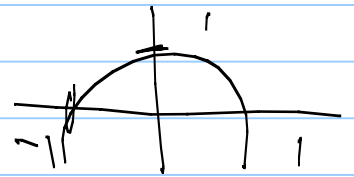
$$\textcircled{1} \quad |2x-1| = \begin{cases} 2x-1 & \text{if } 2x-1 \geq 0 \\ -(2x-1) & \text{if } 2x-1 < 0 \end{cases}$$

$$= \begin{cases} 2x-1 & \text{if } x \geq \frac{1}{2} \\ -(2x-1) & \text{if } x < \frac{1}{2} \end{cases}$$

$$\textcircled{2} |1-x| = \begin{cases} 1-x & \text{if } 1-x \geq 0 \\ -1+x & \text{if } 1-x < 0 \end{cases}$$

$$= \begin{cases} 1-x & \text{if } 1 \geq x \\ -1+x & \text{if } 1 < x \end{cases}$$

$$\begin{aligned}
 \textcircled{3} \quad |1-x^2| &= \begin{cases} 1-x^2 & \text{if } 1-x^2 \geq 0 \\ -1+x^2 & \text{if } 1-x^2 < 0 \end{cases} \\
 &= \begin{cases} 1-x^2 & \text{if } -1 < x < 1 \\ -1+x^2 & \text{if } x < -1 \text{ and } x > 1 \end{cases}
 \end{aligned}$$



Properties of Absolute Value pA7

$$1. |ab| = |a||b| \qquad 3. |a^n| = |a|^n$$

$$2. \left| \frac{a}{b} \right| = \frac{|a|}{|b|} \quad (b \neq 0)$$

$$4. \sqrt{x^2} = |x|$$

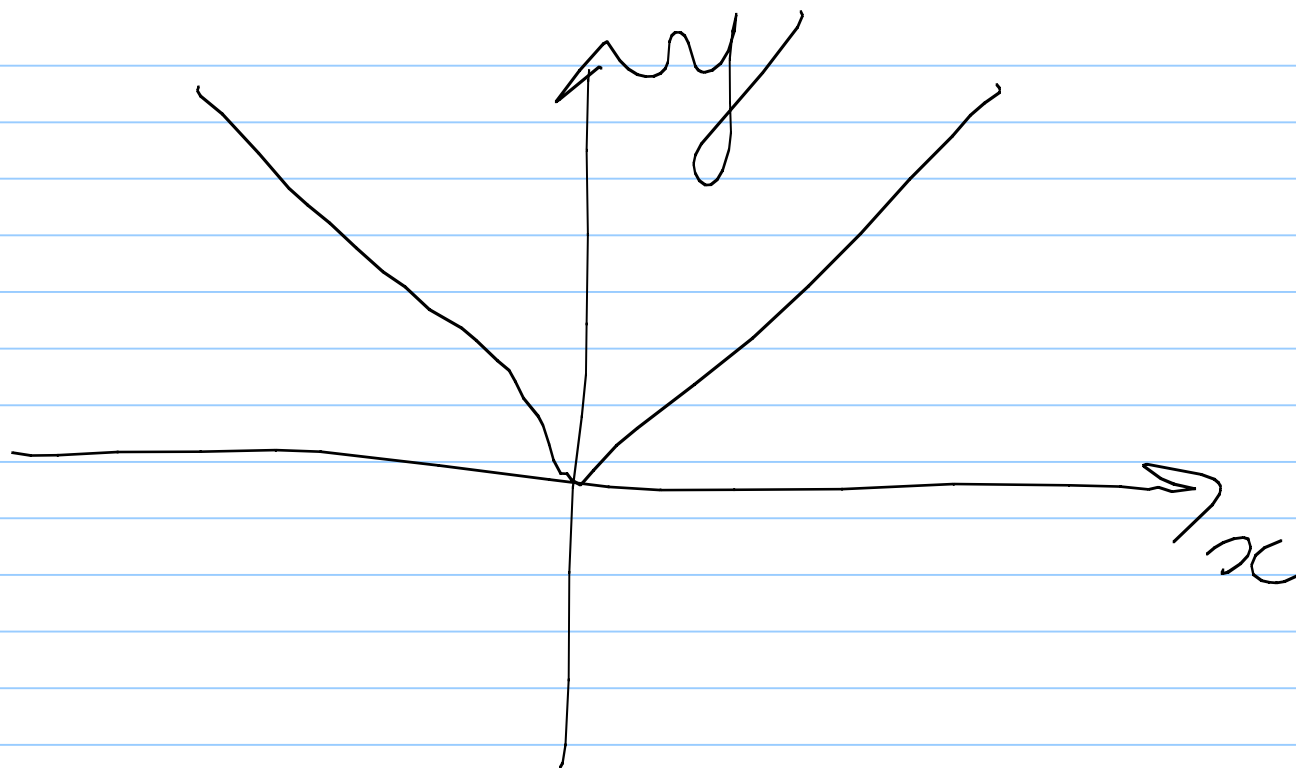
$$5. |x| > 0 \quad \text{for all } x \in \mathbb{R}$$

$$6. |x| = 0 \quad \text{if and only if } x = 0$$

$$7. |x| \leq |y| \quad \text{if and only if } x^2 \leq y^2$$

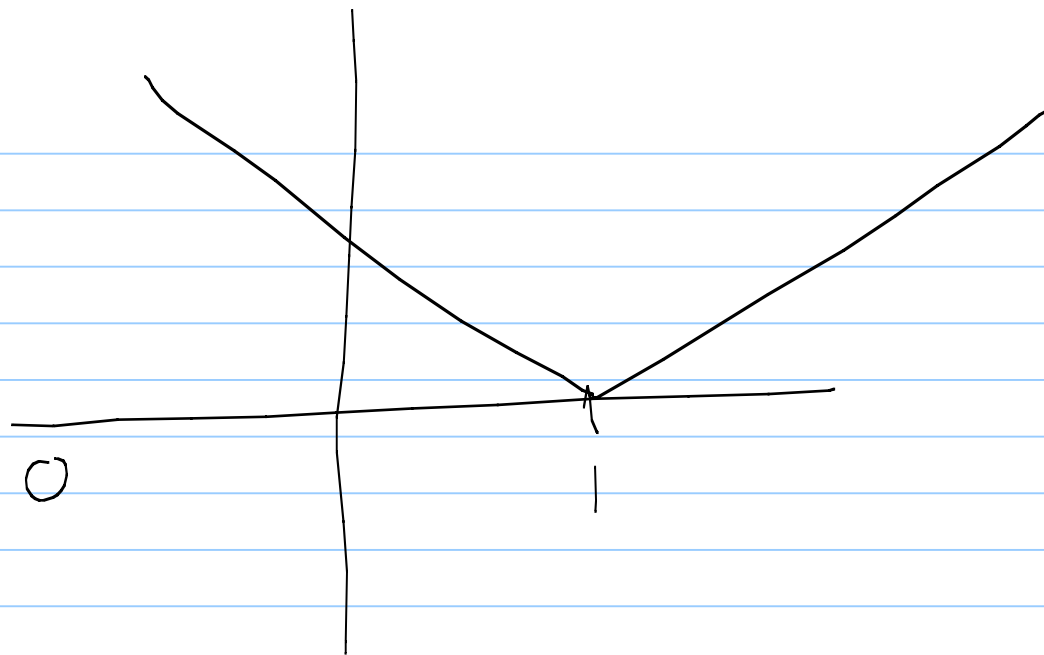
Sketch

① $y = |x|$

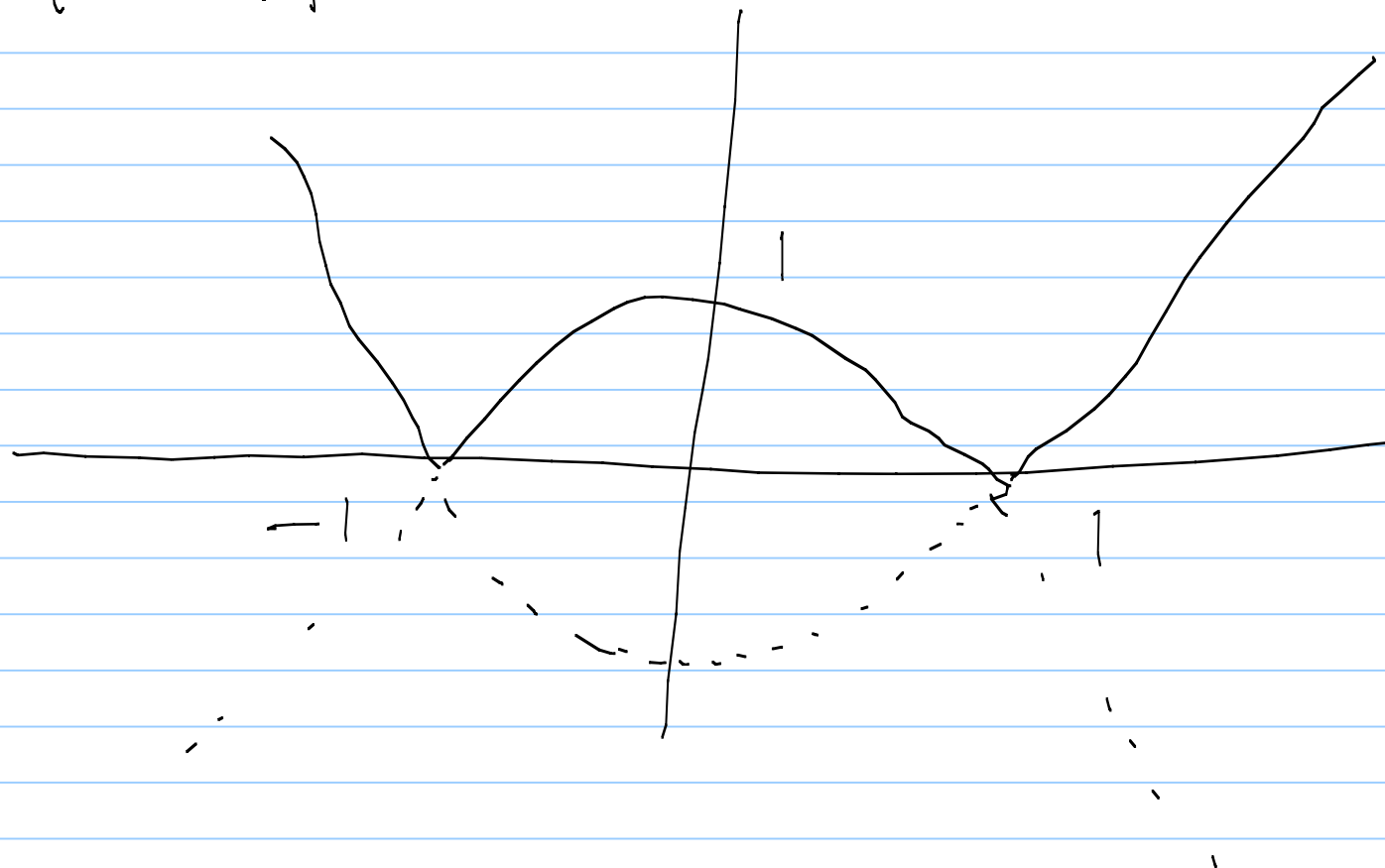


② $y = |x-1|$

$$= \begin{cases} x-1 & \text{if } x-1 \geq 0 \\ -x+1 & \text{if } x-1 < 0 \end{cases}$$

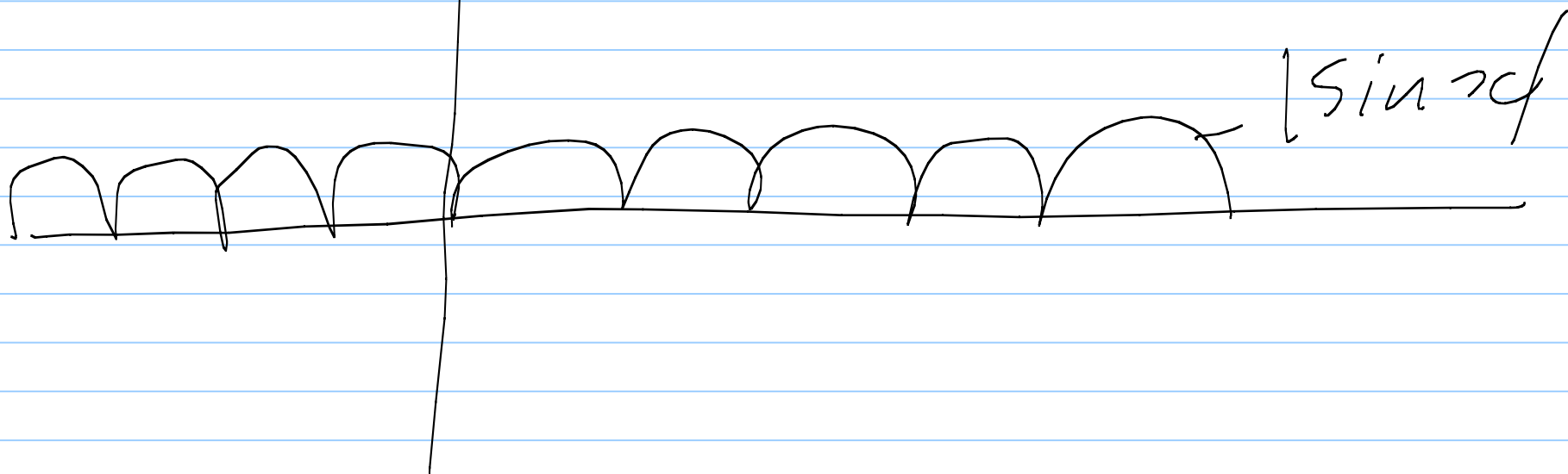
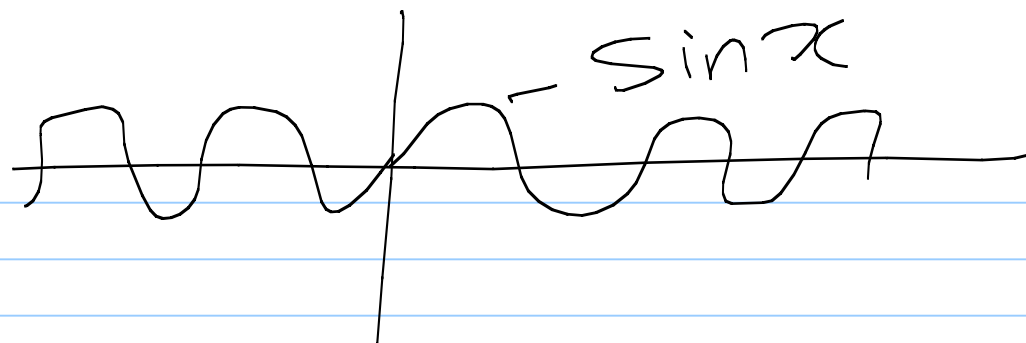


③ $y = |x^2 - 1|$



④

$$y = |\sin x|$$



Solving equalities and inequalities with absolute value

Thm 6 p A7 (very important)

Suppose $a > 0$. Then

1. $|x| = a$ if and only if $x = \pm a$
2. $|x| < a$ if and only if $-a < x < a$
3. $|x| > a$ if and only if $x > a$ or $x < -a$

Example Solve

$$\textcircled{1} |2x - 5| = 9$$

$$2x - 5 = 9$$

OR

$$2x - 5 = -9$$

$$\Rightarrow 2x = 14$$

OR

$$2x = -4$$

$$\Rightarrow x = 7$$

OR

$$x = -2$$

$$\Rightarrow x \in \{-2, 7\}$$

Example

$$\textcircled{1} \quad |x+1| \geq 3$$

$$\Rightarrow x+1 \geq 3 \quad \text{OR} \quad x+1 \leq -3$$

$$\Rightarrow x \geq 2 \quad \text{OR} \quad x \leq -4$$

$$\Rightarrow x \in (-\infty, -4] \cup [2, \infty)$$

$$\textcircled{2} \quad |3 - 2x| < 5$$

$$\Rightarrow -5 < 3 - 2x < 5$$

$$\Rightarrow -8 < -2x < 2$$

$$\Rightarrow 4 > x > -1$$

$$\Rightarrow x \in (-1, 4)$$

$$\textcircled{3} \quad |x^2 - 9| = 1$$

$$\Rightarrow x^2 - 9 = 1 \quad \text{OR} \quad x^2 - 9 = -1$$

$$\Rightarrow x^2 - 10 = 0 \quad \text{OR} \quad x^2 - 8 = 0$$

$$\Rightarrow (x - \sqrt{10})(x + \sqrt{10}) = 0 \quad \text{OR} \quad (x - \sqrt{8})(x + \sqrt{8}) = 0$$

$$\Rightarrow x \in \{-\sqrt{10}, -\sqrt{8}, \sqrt{8}, \sqrt{10}\}$$

$$\textcircled{5} \quad |x-1| < -4 \quad x \in \emptyset$$

$$\textcircled{6} \quad |x-1| > -4 \quad x \in \mathbb{R}$$

$$\textcircled{7} \quad |x| < -x \quad x \in \emptyset$$

$$\textcircled{8} \quad \frac{|1+2x|}{1-x} \leq 0$$

$$\Rightarrow 1+2x = 0 \quad \text{and} \quad 1-x < 0$$

$$\Rightarrow x = -1/2 \quad \text{and} \quad 1 < x$$

$$\Rightarrow x \in (1, \infty)$$

$$\textcircled{a} \quad \frac{-1+x}{|1+2x|} \leq 0$$

$$\Rightarrow 1+2x \neq 0 \text{ and } -1+x \leq 0$$

$$\Rightarrow x \neq -\frac{1}{2} \text{ and } x \leq 1$$

$$x \in (-\infty, -\frac{1}{2}) \cup (-\frac{1}{2}, 1]$$

$$\textcircled{10} \quad \frac{|1-x|}{|1+x|} \geq 0$$

$$\Rightarrow 1+x \neq 0 \quad \text{and} \quad x \neq -1$$

$$\Rightarrow x \in (-\infty, -1) \cup (-1, \infty)$$

$$\{x \mid x \neq -1\}$$

$$x \in \mathbb{R} \setminus -1$$

Examples

$$\textcircled{1} |2x - 5| = 3$$

$$\textcircled{1} |x - 2| > 3$$

$$\textcircled{2} |2 - 4x| \leq 6$$

$$\textcircled{3} \frac{x}{|x - 2|} > 0$$

$$\textcircled{4} \frac{|x|}{|x - 1|} > 1$$

$$\textcircled{5} \frac{|x + 1|}{|x - 3|} > 0$$

$$\textcircled{6} |x^2 - 9| = 14$$

$$\textcircled{7} |x - 4| > 3$$

$$\textcircled{8} \frac{|x - 1|}{x + 1} > 0$$

Sketch: ① $y = |2x + 1|$

$$\textcircled{2} y = |4 - x^2|$$

$$\textcircled{1} \quad |2x - 5| = 3$$

$$\Rightarrow 2x - 5 = 3 \quad \text{OR} \quad 2x - 5 = -3$$

$$\Rightarrow 2x = 8 \quad \text{OR} \quad 2x = -2$$

$$\Rightarrow x = 4 \quad \text{OR} \quad x = -1$$

$$\textcircled{1} \quad |x-2| > 3$$

$$\Rightarrow x-2 > 3 \quad \text{OR} \quad x-2 < -3$$

$$\Rightarrow x-2 > 3 \quad \text{OR} \quad x-2 < -3$$

$$\Rightarrow x > 5 \quad \text{OR} \quad x < -1$$

$$\Rightarrow x \in (-\infty, -1) \cup (5, \infty)$$

$$\textcircled{2} \quad |2-4x| \leq 6$$

$$\Rightarrow -6 \leq 2-4x \leq 6$$

$$\Rightarrow -8 \leq -4x \leq 4$$

$$\Rightarrow 2 \geq x \geq -1$$

$$\Rightarrow x \in [-1, 2]$$

$$\textcircled{3} \quad \frac{x}{|x-2|} > 0$$

$$\Rightarrow x > 0 \quad \text{and} \quad x-2 \neq 0$$

$$\Rightarrow x \in (0, 2) \cup (2, \infty)$$

$$\textcircled{4} \quad \frac{|x|}{|x-1|} > 0$$

$$\Rightarrow x-1 \neq 0$$

$$\Rightarrow x \neq 1$$

$$\Rightarrow x \in (-\infty, 1) \cup (1, \infty)$$

$$\textcircled{5} \quad \frac{|x+1|}{|x-3|} > 0$$

$$\Rightarrow x-3 \neq 0 \text{ and } x+1 \neq 0$$

$$\Rightarrow x \neq 3 \text{ and } x \neq -1$$

$$\Rightarrow x \in (-\infty, -1) \cup (-1, 3) \cup (3, \infty)$$

$$\textcircled{6} \quad |x^2 - 9| = 14$$

$$\Rightarrow x^2 - 9 = 14 \quad \text{OR} \quad x^2 - 9 = -14$$

$$\Rightarrow x^2 - 23 = 0 \quad \text{OR} \quad x^2 + 5 = 0$$

$$\Rightarrow (x - \sqrt{23})(x + \sqrt{23}) = 0 \quad \text{OR} \quad x \in \emptyset$$

$$\Rightarrow x \in \{-\sqrt{23}; \sqrt{23}\}$$

$$\textcircled{7} \quad | \quad x-4 \quad | \quad > \quad 3$$

$$\Rightarrow x-4 > 3 \quad \text{OR} \quad x-4 < -3$$

$$\Rightarrow x > 7 \quad \text{OR} \quad x < 1$$

$$\Rightarrow x \in (-\infty, 1) \cup (7, \infty)$$

$$\textcircled{8} \quad \frac{|x-1|}{x+1} > 0$$

$$\Rightarrow x-1 \neq 0 \quad \text{and} \quad x+1 > 0$$

$$\Rightarrow x \neq 1 \quad \text{and} \quad x > -1$$

$$\Rightarrow x \in (-1, 1) \cup (1, \infty)$$